

Azure Cloud Manual Setup Guide - RetailSense AI

This guide provides step-by-step instructions on how to manually set up and deploy the entire RetailSense AI cloud architecture using both the **Azure Portal** and **Azure CLI**.

■ Step 1: Create a Resource Group

A Resource Group is a logical container that holds related Azure resources.

Option A: Azure Portal

1. Navigate to the Azure Portal.
2. Search for **Resource groups** and click **Create**.
3. Select your subscription, name the resource group `rg-retailsense`, and select a region (e.g., `East US`).
4. Click **Review + create** -> **Create**.

Option B: Azure CLI

```
az group create --name rg-retailsense --location eastus
```

■ Step 2: Create a Storage Account (ADLS Gen2)

We need a storage layer to hold our raw, Bronze, Silver, and Gold Parquet/Delta files. For high-performance big data operations, we use **Azure Data Lake Storage Gen2 (ADLS Gen2)**, which enables **Hierarchical Namespace**.

Option A: Azure Portal

1. Search for **Storage accounts** -> Click **Create**.
2. Set **Resource group** to `rg-retailsense` and name the account (e.g., `stretailsensedata`).
3. Select **Standard** performance and **Locally-redundant storage (LRS)** to minimize costs.
4. Go to the **Advanced** tab, scroll down to **Data Lake Storage Gen2**, and check **Enable hierarchical namespace**.
5. Click **Review + create** -> **Create**.
6. Once deployed, open the resource, go to **Containers**, and create a container named `data`.

Option B: Azure CLI

```
# Create storage account with ADLS Gen2 enabled (--enable-hierarchical-namespace true)<br/>az storage account create \<br/> --name stretailsensedata \<br/> --resource-group rg-retailsense \<br/> --location eastus \<br/> --sku Standard_LRS \<br/> --kind StorageV2 \<br/> --enable-hierarchical-namespace true<br/><br/># Create 'data' container<br/>az storage container create \<br/> --name data \<br/> --account-name stretailsensedata
```

■ Step 3: Create an Azure Databricks Workspace

Databricks provides the managed Apache Spark cluster environment.

Option A: Azure Portal

1. Search for **Azure Databricks** -> Click **Create**.
2. Set **Resource group** to **rg-retailsense** and workspace name to **db-retailsense**.
3. Set **Pricing Tier** to **Premium** (required for Unity Catalog and native Notebook Dashboards).
4. Click **Review + create** -> **Create** (takes 2-4 minutes).

Option B: Azure CLI

```
az databricks workspace create \<br/> --resource-group rg-retailsense \<br/> --name db-retailsense \<br/> --location eastus \<br/> --sku premium
```

■ Step 4: Mount Storage to Databricks (DBFS)

Mounting allows PySpark to interact with ADLS Gen2 as if it were a local directory.

1. Open your Databricks Workspace.
2. Create a temporary notebook and run the following Python code to mount the storage account:

```
# Configure credentials using Storage Account SAS Token or Account Key<br/>dbutils.fs.mount(<br/> source = "wasbs://data@stretailsensedata.blob.core.windows.net",<br/> mount_point = "/mnt/data",<br/> extra_configs = {"fs.azure.account.key.stretailsensedata.blob.core.windows.net": "<YOUR_STORAGE_ACCOUNT_KEY>"}<br/>)
```

■ Step 5: Link the GitHub Repository

Link your git repository to keep Databricks code synchronized.

1. In Databricks, click on **Workspace** in the sidebar -> **Repos** (or **Git Folders**).
 2. Click **Add Repo** -> Paste the Repository URL: <https://github.com/KISHANSINHAA/retail-m5.git>.
 3. Databricks will pull your code into the workspace.
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■■ Step 6: Configure Environment & Libraries

Set up cluster configurations to support execution.

1. Navigate to **Compute** -> **Create compute** (Select **Serverless** or create a **Single Node** cluster).
2. If using a provisioned cluster, go to the **Libraries** tab and install:
 - **xgboost** (via PyPI)
 - **scikit-learn** (via PyPI)
 - **pydantic-settings** (via PyPI)
 - **python-dotenv** (via PyPI)
3. Set cluster environment variables to target your mounted cloud storage:

```
RAW_DIR=/dbfs/mnt/data/raw<br/>    BRONZE_DIR=/dbfs/mnt/data/bronze<br/>    SILVER_DIR=/dbfs/mnt/data/silver<br/>    GOLD_DIR=/dbfs/mnt/data/gold<br/>    MODELS_DIR=/dbfs/mnt/data/models<br/>    SAMPLE_LIMIT=0
```